A Low Cost, Efficient Electrical Power System Design for CubeSat

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***Abstract* — This paper discusses the design of an Electrical Power System (EPS) for a 1U CubeSat. A low-cost, lightweight, and highly efficient EPS using commercial off-the-shelf (COTS) components is designed. The EPS comprises of three modules: Power generation, Power storage, and Power distribution. To determine the amount of power CubeSat would receive, various analyses are conducted on eclipse time and inclination. A polar orbit is chosen to minimize eclipse time without expending extra power to maintain the satellite in a retrograde orbit. Polycrystalline silicon solar panels are selected due to their decent efficiency, affordability, and availability. LG MJ1 batteries are chosen for their high specific energy, high maximum continuous discharge, stable end of discharge, among other factors. A detailed power budgeting and cost analysis of the proposed EPS is also conducted. The analysis shows that the chosen batteries can provide 3.60 W of power, sufficient to meet the required power of 3.55 W in the worst-case scenario for the satellite, with an expected total cost of $109.73.**

Keywords— cubesat, electrical power system (EPS), low-earth-orbit (LEO) satellites, multi-junction (MJ) solar cells, maximum power point tracking (MPPT), power generation, power budgeting

I. Introduction

A CubeSat is a nanosatellite with dimensions of 10×10×10 cm and a mass of 1.33 kg. Like other satellites, the main source of energy for a CubeSat is solar energy [1]. The Electrical Power System (EPS) of a CubeSat collects solar energy, converts it to electrical energy, stores it, and distributes it to the satellite's other subsystems. The EPS is vital to the satellite and must be built to ensure maximum efficiency and robust performance [2]. All parts of the CubeSat, including the entire satellite's subsystems, must be powered by the EPS. Throughout the eclipse time, the EPS must supply sufficient power.

The CubeSat in low-earth orbit (LEO) takes around 90 minutes to orbit, completing 16 orbits per day, with 35 minutes and 45 seconds in eclipse and 58 minutes and 52 seconds in sunlight. Without power generation, the EPS must store enough power to support satellite functioning for the entire 35 minutes and 45 seconds eclipse. Various studies have focused on different aspects of CubeSat EPS design, such as power generation estimates, addressing latency and capacity needs for 5G applications, and distributed EPS designs for CubeSat or Nano class satellites [1, 3, 4].

Previous works have placed their 1-unit CubeSats in Low Earth Orbit (LEO) with different orientations and orbits, such as Sun Synchronous Orbit (SSO) or orbits similar to the International Space Station (ISS) [5-6]. However, there is room for improvement in designing an efficient and cost-effective EPS for CubeSats.

This paper proposes an EPS for a 1-unit CubeSat located in Low Earth Orbit (LEO) with a 90° inclination, intending to include 7 subsystems. The design aims to be low-cost, highly efficient, and capable of providing the required power for all subsystems. The worst-case scenario, in which all subsystems are in power mode simultaneously, is considered to ensure sufficient power supply. This can be achieved using optimized components, designs, and algorithms. The CubeSat's orientation is chosen as Nadir-Pointing due to its ability to generate maximum power on three sides (X+, Z+, and Z-). Additionally, the X- side gains power during the phases just prior and following the eclipse, maximizing power generation for the CubeSat.

II. Orbit Inclination Eclipse Time

Calculating eclipse time is crucial for understanding the amount of solar energy that solar panels receive in orbit and

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for subsequent power budgeting. Eclipse time decreases as inclination increases [7]. This is because, with an increase in inclination, the angle β also increases, leading to greater solar illumination and a consequent decrease in eclipse time.

Placing the satellite with an inclination i such that $90° \leq i \leq 120°$ decreases the mean eclipse time. The selected inclination for this study is 90°. Satellites in a retrograde orbit ($90° < i \leq 180°$) require more fuel for launch, which is why most satellites are not placed in such orbits, especially with values significantly greater than 90°.

Additionally, a larger β value could result in an overall increase in satellite temperature, negatively affecting the efficiency of solar cells and battery life [4].

A. Calculation of Time in Eclipse and Sunlight

The CubeSat in LEO takes around 90 minutes to orbit, completing 16 orbits per day. Based on several research papers, the eclipse time and the sunlight time have been calculated as follows [7].

The radius of the Earth (R_{earth}) is 6,378 km, and h represents the altitude of the CubeSat, which can range from 400 km to 600 km. In this work, the altitude is selected to be 500 km.

$$a = R_{earth} + h = 6378 + 500 = 6878\ km \quad (1)$$

The orbit velocity and the period are calculated as:

$$\vartheta = \sqrt{\frac{\mu}{a}} = \sqrt{\frac{3.986\times10^{\wedge}5}{6878}} = 7.61\text{km/s} \quad (2)$$

$where, \mu = GM$, is the product of the gravitational constant (G) and the mass of the Earth (M), has a value of 3.986×10^5 km³/s².

and the period is calculated as:

$$P = 2\pi\sqrt{\frac{a^i}{\mu}},\ where\ i = 3$$

$$P = 2\pi \times \sqrt{\frac{a^3}{\mu}} = 2\pi \times \sqrt{\frac{6878^3}{3.986\times10^5}} = 5676.81 \quad (3)$$

Earth's angular radius at mission altitude is given by $\rho = \sin^i R/a$ where $i = -1$. Hence,

$$\rho = \sin^{-1}\left(\frac{R}{a}\right) = \sin^{-1}\left(\frac{6378}{6878}\right) = 68.02^{\circ} \quad (4)$$

The maximum time of the eclipse (TE) and the time in sunlight (TS) are given by:

$$TE = \frac{2\rho}{360} \times P \quad (5)$$

$$\left(\frac{2 \text{ x } 68.02}{360}\right) \times 5676.812 = 2145.2\ s = 35.75\ min$$

$$TS = P - TE \quad (6)$$

$$5676.812 - 2145.2 = 3531.612\ s = 58\ min\ 52\ s$$

Obtained data are summarized in Table I.

TABLE I. SPECIFICATION OF EPS

Orbital Parameter	Requirement
Altitude	500 km
Inclination	90 degrees
Period	94 min and 37 s
Eclipse	35 min 45 s
Sunlight	58 min 52 s
Spacecraft Size	1U
Number of orbits per day	16

III. PROPOSED TOPOLOGY FOR ELECTRICAL POWER SUBSYSTEM DESIGN

Being a vital component, the EPS needs to be both adaptable and efficient in order for the CubeSat to become a standard vehicle utilized for missions in the real world.

The Electrical Power Subsystem (EPS) has the responsibility of generating power by the use of solar panels, storing the power using batteries, and distributing it using the power board. The main responsibilities of the power system are generating adequate power, and assuring effective battery recharge, while following the physical constraints of the CubeSat. EPS has mainly three components which are power generation, power storage, and power distribution. Each of them consists of various components such as converters, regulators, and batteries. Fig. 1 illustrates an example configuration and the circuit connection of the EPS system.

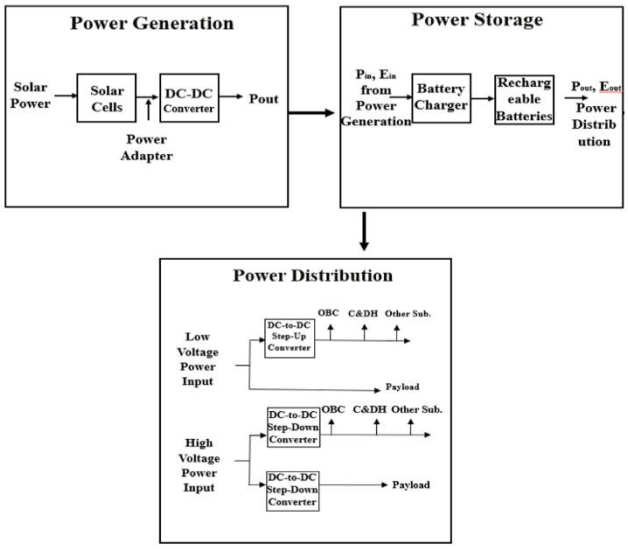

Fig. 1. EPS circuit block diagram.

A. Power Generation

The design of the power generation part includes solar cells, a power adapter, and a DC-to-DC converter. Since the primary goal is to maintain high efficiency at a very low cost, polycrystalline solar panels are chosen due to their low cost and sufficient power output.

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These panels have a peak current of 60 mA and a peak voltage of over 5 volts in maximum sunlight. They are lightweight, inexpensive, small, and offer an efficiency between 21.50-25.30%. The output power of the solar panels is DC power. The power subsystem's role is to supply DC power to all subsystems for the duration of the mission. The power adapter is a voltage converter that performs rectification, voltage regulation, and voltage transformation with a DC output. The DC-DC converter is used to deliver the voltage as required by the subsystem. If the output of the DC-to-DC Converter is lower than what the load requires, then the step-up converter steps up the voltage to that required by the subsystem.

B. Power Storage

The power storage component receives power and energy from the power generation component, which is then output to the battery charger. As shown in Fig. 1, the battery charger is often connected directly to the battery, which in turn is linked to the system load. Furthermore, there are battery chargers available that can simultaneously support a load and charge the batteries.

Battery performance is a critical aspect of this work. Therefore, the focus has been placed on maintaining and ensuring the efficiency and life of the batteries for the entire mission duration. As the battery charges and discharges, it will display a sinusoidal graph. However, the most important aspect is to ensure that the battery has a positive linear relationship between the charging percentage and time. The battery should continue charging and increasing the percentage of charge over time. The most commonly used type of lithium-ion cells for CubeSats is the cylindrical 18650 COTS cells. The LG MJ1 is chosen because it is a commercial cell that has been used for various nanosatellites by space agencies like NASA. Krause et al compared the performance of LGMJ1 to other COTS cells. It outperformed the others in terms of specific energy, life cycle, and capacity along with a stable end-of-discharge voltage [8]. The cell we chose has a nominal capacity of 3500mAh, a continuous discharge rate of 10A, a nominal voltage of 3.6V, and weighs 46.90g. Since one of the important things that must be ensured in this project is to provide a sufficient amount of energy and to ensure the performance of the battery, so we used two batteries. The batteries are light in weight, and they will not affect the CubeSat efficiency.

C. Power Distribution

A step-up device is necessary for power distribution because the first option that is shown in Fig. 1 assumes a low input voltage from the batteries. Step-down devices are required when a higher input voltage is present. Based on [9], the chosen step-up/down device is a DC-to-DC converter.

IV. Power Budgeting

The power budget is determined by the amount of power needed by each subsystem, the amount of power produced by solar cells, and the amount of power stored in batteries. The power budget will be positive if the power generated during one orbit is greater than or equal to the power used during the orbit. In contrast, it will be negative if the power generated during one orbit is less than the power spent during this orbit; in this scenario, the battery will steadily drain until it is empty. In the proposed work, the payload was set to be a CMOS Camera, which is a type of imaging equipment. It is also a type of low-power sensor with a monochrome image, suitable for applications such as tracking, and objects. Due to its low cost and minimum power consumption, it has been selected as the payload of this work. This payload has a power consumption of 70 mW and costs roughly $14.

Based on the selected payload, the duration for each subsystem in our CubeSat has been calculated, with the final values shown in Table II and the power needed for each power mode shown in Table III. The following equation was used to calculate the energy [10].

$$E_e = P_{nd} \times T_e \tag{7}$$

TABLE II: POWER NEEDED, DURATION, AND ENERGY CONSUMPTION FOR EACH SUBSYSTEM

Subsystem	Power Consumed[mW]	Duration [h]	Energy [mWh]
EPS	432	0.98	423
Battery Heater	272	0.79	215
OBC	130	0.98	127
COM-RX	180	0.79	142
COM-TX	56	0.13	7
Payload	70	0.17	12
Antenna (Nominal)	20	0.98	20
Antenna (Deployment)	2000	0.98	1960
Solar Panel	10	0.98	10
Thermal	140	0.79	111
C&DH	140	0.79	111
ADCS	100	0.79	79
Total	**3550**		**3217**

TABLE III: POWER NEEDED FOR EACH SUBSYSTEM IN DIFFERENT MODES

	EPS mW	OBC mW	COM -RX mW	COM -TX mW	Antennas mW	Payload mW	C&DH mW	Thermal mW	ADCS mW	Total Power mW
Separation mode	442	130	—	—	2020	—	—	—	—	2592
Initialization mode	442	130	—	—	2020	—	—	—	—	2592
Safe mode	442	130	—	—	—	—	—	—	—	572
Nominal mode T-P	442	130	180	56	2020	70	140	412	100	3550
Nominal mode T- np	442	130	180	56	2020	—	140	412	100	3480
Nominal mode nT-P	442	130	180	—	2020	70	140	412	100	3492
Nominal mode nT-nP	442	130	180	—	2020	—	140	412	100	3424

From the battery specification, the special energy is 266 (Wh/kg) and the weight of one battery is 0.05 kg. From these data, the value of the supplied energy in (Wh) was calculated as 266 Wh/kg; Therefore, $Wh = 13.03$ for one battery and 26.07 for two batteries.

Comparing this supplied energy with the energy consumption for the CubeSat of 3.22 Wh, it is clear that the

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batteries will fulfil the requirement and will supply our CubeSat with more than the power needed.

A. Power Generated by Polycrystalline Solar Panels

Provided that our 1U CubeSat has fixed Polycrystalline silicon solar panels in Low Earth Orbit (LEO) with a 90° inclination. The CubeSat's orientation is always Nadir-Pointing. Solar panels are mounted on 3 sides, and the 4th side is always pointing to Nadir. To calculate the total power output for this CubeSat in watts per orbit, we need to consider several factors.

- First, in this work, the CubeSat has dimensions of 10 cm x 10 cm x 10 cm, and the solar panels cover three sides with a total surface area of 300 cm² (3 x 10 cm x 10 cm).
- Second, the chosen Polycrystalline solar panels typically have an efficiency of around 21.5 - 23.5 %. We will use an efficiency of 22 % for our calculations.
- Third, the solar irradiance outside Earth's atmosphere (at 1 AU from the Sun) is about 1366 W/m².

Because the CubeSat is nadir-pointing, the three sides with solar panels will not receive equal sunlight throughout the orbit. The side facing the Sun will receive the most sunlight, while the other two sides will receive less due to the angle between the panels and the Sun. We'll assume a cosine loss factor to account for this. In our calculations, we'll use 1 for the fully sunlit side, and 0.5 for the other two sides (since they are not optimally aligned with the Sun).

Calculating the power generation as follow:

Power (W) = Solar Irradiance (W/m²) x Solar Panel Area (m²) x Solar Panel Efficiency x Cosine Loss Factor (8)

First, convert the solar panel area to square meters: 300 cm² = 0.03 m²

Calculating the power generation for each side separately, considering the cosine loss factor:

Power_side1 (W) = 1366 W/m² x 0.01 m² x 0.22 x 1 ≈ 3 W
Power_side2 (W) = 1366 W/m² x 0.01 m² x 0.22 x 0.5 ≈ 1.5 W
Power_side3 (W) = 1366 W/m² x 0.01 m² x 0.22 x 0.5 ≈ 1.5 W

The total power is:

Total Power (W) = Power_side1 (W) + Power_side2 (W) + Power_side3 (W)
Total Power (W) ≈ 3 W + 1.5 W + 1.5 W ≈ 6 W

This is the average power generation when the CubeSat is in sunlight. Assuming the CubeSat spends about 60% of its orbit in sunlight, the average power generation over the entire orbit can be calculated as the following:

Average Power (W) = Total Power (W) x Sunlight Percentage (9)
Average Power (W) ≈ 6 W x 60% ≈ 3.60 W.

As a final result, the power needed is 3.55 W and the power generated from the solar panels is 3.60 W. We can notice that the system now can fulfill the power needed for the CubeSat as the minimum power generated is greater than the power consumed or needed for this CubeSat.

$$P_{generated} > P_{consumed}$$

3.60 W > 3.55 W

This is for the minimum power generated from the solar panels and in the worst case when all subsystems are active since it is unlikely for all subsystems to be ON throughout CubeSat's lifetime.

B. Energy Used

The energy used during the eclipse time by using the following equation:

$$E_e = P_{nd} \times T_e \quad (10)$$

where P_{nd} represents the power available to the subsystem including loss, and T_e is the Eclipse Time.

$$E_e = 3.6\,W \times 0.6\,h = 2.16 \text{ Wh}$$

The required minimum capacity of the battery for the nanosatellite is another crucial value to be aware of. The nominal voltage V of our battery is 3.64 V, where the energy used during the eclipse is approximately 2.16 Wh, and the batteries' stringent minimum capacity is:

$$C = \frac{E_e}{V} \quad (11)$$

$$\frac{2.16\,wh}{3.64\,V} = 0.59\,Ah = 590\,mAh.$$

The capacity of Li-ion batteries slightly decreases at each charge and discharge cycle. A depth of discharge (DoD) of 30% or less is recommended for this kind of application. The ideal capacity for our battery is:

$$C_i = \frac{C}{DoD} \quad (12)$$

$$\frac{590\ mAh}{0.30} \approx 1967\,mAh$$

As a result of the final calculations, this nanosatellite needs batteries with a minimum capacity of 1967 mAh to complete this mission. Table IV summarizes the final specifications.

TABLE IV: FINAL SYSTEM SPECIFICATIONS

Orbital Parameter	Requirement
Altitude	500 km
Spacecraft Size	1U
Inclination	90 degrees
Number of orbits per day	16
Period	94 minutes and 37 seconds
Eclipse	35 min and 45 seconds
Sunlight	58 min and 52 seconds
Power Needed	3.55 W
Average power Generated in sunlight time	3.6 W
Mission Time	1 Year
Total Energy Required	3.24 Wh
Energy Supplied By 1 Battery	13.03 Wh

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TABLE V: COMPARISON BETWEEN OUR PROPOSED WORK AND OTHER PAPERS

Ref.	Cost	Unit	Payload	Solar Panel	Number of Subsystems	Power Modes	Inclination	Altitude	Calculations	Type of Orbit	Type of orientation	Sunlight Time	Eclipse Time
[3]	NA	1U	Communication System	MJ solar panel *$429 per piece*	3	NA	90°	NA	Few	LEO	Nadir-orientation scenario	56.36 minutes	36.04 minutes
[5]	$1600	1U	Imaging Camera [Between $136.14- $272.28]	Azur Space cells $980 for all	5	3	NA	500 Km	NA	Sun Synchronous Orbit (SSO)	NA	59.55 minutes	35.17 minutes
[7]	NA	1U	Communication System	MJ solar panel *$429 per piece*	3	NA	NA	400 Km	Few	LEO	Nadir-orientation scenario	NA	NA
[12]	NA	1U	NA	Triple-junction cells *$429 per piece*	5	7	90°	500 Km	NA	LEO	Nadir-orientation scenario	59.11 minutes	35.5 minutes
[13]	$1,948.57	1U	GPS Payload $100.00	Spectro lab [Dual Junction] $1,100.00	5	NA	NA	400 Km	Few	Sun Synchronous Orbit (SSO)	NA	55 min and 32 sec.	37.24 min
[14]	NA	2U	INMS (Ion/Neutral Mass Spectrometer), which will measure the mass of ions and neutral atoms. [Very expensive]	P110 series solar panels $89.58 per piece	5	3	NA	NA	NA	LEO	Sun Pointing	NA	NA
[15]	More than $500	1U	DC/ Plasma Impedance Probe Payload [892 g- Required Power 870 mW] *GPS Scintillation* [892 g, Required Power 1150 mW]	NA	5	NA	60°	300 Km	NA	LEO	Random Orientation	NA	NA
Proposed Design	$109.73	1U	CMOS camera payload [27.21554 g mass -70 mW of power consumed] $13.22	Polycrystalline Solar Panels Cells [112.83 grams for all panels] $39.75	7	7 modes Considering the worst-case scenario	90°	500 Km	Deep	LEO	Nadir-orientation scenario	58.8602 min=0.981h	35.75 min =0.596h

High-efficiency solar panels with low cost have been used to meet the needs of CubeSat since all the subsystems were included, which is not available in other works as shown in Table V below.

In addition, the worst-case scenario was taken into account when all the subsystems are in operating mode, which is unlikely, but nevertheless, it was validated that the power subsystem can provide sufficient power for all the subsystems. In terms of weight, as the 1-unit CubeSat should be equal to or less than 1.33 kg, we have made our efforts to choose components that provide high efficiency with less weight.

Table VI shows the cost and weight for each component, and we can see that the total weight for the CubeSat is around 576.83 g, thus achieving a low weight for this work. Now we can say that this proposed work achieves an affordable, efficient, and lightweight EPS for CubeSat with all subsystems.

TABLE VI: COST AND WEIGHT OF THE COMPONENTS

Components	Cost (USD)	Weight (g)
SUNYIMA mini polycrystalline solar panels	19.06	112.83
2 Batteries	6.81	93.78
LCD Infrared Obstacle Sensor	4.63	20
Arduino Uno R3 ATmega328P with USB Cable	13.61	90
INA219 DC Current Power Supply Sensor Breakout Board Module Sensor Module I2C interface For Arduino DIY DC INA219B	8.98	9.1
lot Irf9540 MOSFET P-Ch 100V 23A To-220Ab Irf9540npbf	34.85	6.42
2N2222A Transistor	8.17	3.8 each
LM 2596 DC to DC Buck Converter 3.0-40V to 1.5-35V Power Supply Step Down Module	5.45	40.9
Resistors- diodes-capacitors- inductors – transistors - wires	8.17	200
Total	***109.73***	***576.83***

V. CONCLUSION

In this work, we have proposed a highly efficient, low-cost, and lightweight EPS system for a 1-unit CubeSat. It has been analytically shown that the EPS subsystem will generate enough power and ensure that the CubeSat can sustain itself after being launched into space. Components such as batteries and other factors have been taken into consideration to satisfy the requirements of this work. Creating a Flat Sat enables us to thoroughly study the satellite's operations before its launch, helping to detect faults, reduce costs, and minimize space traffic by decreasing the number of failed satellite missions sent to space. The power budgeting conducted in this work can serve as a reference point for others when designing a 1U CubeSat.

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